

AB RENEWABLE ENERGY ENGINEERING

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DIRECT COMBUSTION

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1. Solid carbonaceous material derived from a living plant matter.

- a. Petroleum fuel
- b. Biomass fuel
- c. Fuel cell
- d. None of the above

2. Biomass has generally lower calorific value of ____.

- a. less than 5,000 kcal/kg
- b. 5,000 to 10,000 kcal/kg
- c. 10,000 to 15,000 kcal/kg
- d. All of the above

3. Solid-biomass-fuel burning device used for domestic cooking, water heating, smoking, roasting and simmering.

- a. Furnace
- b. Cookstove
- c. Biomass oven
- d. None of the above

4. Percentage primary air requirement for combustion of fuel in cookstoves based on the total air.

- a. 80 %
- b. 60 %
- c. 40 %
- d. None of the above

5. Open-fire stove has a thermal efficiency of ____.

- a. 3-11%
- b. 11-20%
- c. 20-30%
- d. All of the above

6. Thermal efficiency of cookstoves can be increased by ____.

- a. increasing the amount of fuel
- b. using more air for the fuel
- c. insulating the stove
- d. All of the above

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7. A rice husk stove boils 2 liters of water and subsequently evaporated 0.5 liter. The initial temperature of water is 27°C and the amount of fuel consumed in boiling and evaporating water is 1.5 kg. What is the thermal efficiency of the stove? Assume a 540 kcal/kg heat of vaporization of water and 3,000 kcal per kg heating value of fuel.

- a. 5.2 %
- b. 9.2 %
- c. 12.5 %
- d. None of the above

8. Thermal efficiency of a 1.2kW biomass stove is 15%. If wood with 4,000 kcal per kg heating value is used as fuel, what would be its consumption of wood per hour?

- a. 1.6 kg per hour
- b. 2.0 kg per hour
- c. 2.4 kg per hour
- d. None of the above

9. Part of a stove or of a furnace where combustion of fuel takes place.

- a. Chimney
- b. Damper
- c. Firebox
- d. All of the above

10. Recommended angle for the inclined metal grate in a step-grate-type stove.

- a. 45-60 deg
- b. 30-40 deg
- c. 60-80 deg
- d. None of the above

11. Indicator of the amount of fuel in a direct stove that is converted into ash.

- a. Burning efficiency
- b. Thermal efficiency
- c. Heat utilization efficiency
- d. None of the above

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12. Recommended angle for the inclined metal grate in a step-grate type stove.

- a. 45-60 deg
- b. 30-40 deg
- c. 60-80 deg
- d. None of the above

13. Power output of an agriwaste-fired multiple-burner stove can be increased by ____.

- a. providing a large connecting tunnel for the pot holes
- b. providing a damper to control the flow of the passage of air
- c. increasing the height of the chimney
- d. None of the above

14. Heat energy content of coconut shell charcoal is 14223 BTU/lb. How many kilo calories are there in 10 kilos of coconut shell charcoal.

- a. 78, 852.3 Kcal
- b. 68, 852.3 Kcal
- c. 87, 852.3 Kcal
- d. None of the above

15. Fuel blocks produced through simple compaction of biomass, usually plants or plant residues in combination with binding material.

- a. Densified briquettes
- b. Char briquettes
- c. Charcoal briquettes
- d. None of the above

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16. Solid carbonaceous material derived from a living plant matter.

- a. Petroleum fuel
- b. Biomass fuel
- c. Fuel cell
- d. None of the above

17. Biomass have generally lower calorific value of

- a. less than 5,000 kcal/kg
- b. 5,000 to 10,000 kcal/kg
- c. 10,000 to 15,000 kcal/kg
- d. All of the above

18. A bin has a dimension 3m-wide by 4m-long and 6m-high. How much rice husks (100 kg/m³) is required to have it completely filled?

- a. 3.1 tons
- b. 5.4 tons
- c. 7.2 tons
- d. None of the above

19. Which of the following is not a biomass?

- a. Corn cobs
- b. Rice husks
- c. Peat
- d. None of the above

20. Which of the following is not a thermo-chemical conversion process of biomass?

- a. Direct combustion
- b. Pyrolysis
- c. Gasification
- d. None of the above

21. Indicator of the amount of fuel in a furnace that is converted into ash.

- a. Burning efficiency
- b. Thermal efficiency
- c. Heat utilization efficiency
- d. None of the above

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22. Performance efficiency of a furnace based on the energy losses at its transmission, on its walls and at its openings.

- a. Furnace efficiency
- b. Burning efficiency
- c. Heat transfer efficiency
- d. None of the above

23. If a furnace uses biomass with 4,000 kcal/kg heating value, how much biomass is consumed per hour? Assume a 40% furnace efficiency.

- a. 1.25 kg per hour
- b. 2.46 kg per hour
- c. 2.89 kg per hour
- d. None of the above

24. Indicator of the ratio of the amount of heat given off by the furnace to the system and the heat used from the fuel.

- a. Burning efficiency
- b. Thermal efficiency

- c. Heat utilization energy
- d. None of the above
- e. above

25. Performance efficiency of a furnace based on the amount of combustible material present after burning.

- a. Furnace efficiency
- b. Burning efficiency
- c. Heat transfer efficiency
- d. None of the above

26. Thermal efficiency of a furnace is the ratio of ____.

- a. the amount of fuel used to the amount of heat generated during combustion
- b. the amount of heat generated during combustion to the amount of fuel used
- c. the amount of heat delivered to the amount of heat available in the fuel used
- d. None of the above

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27. Thermal efficiency of rice hull furnaces is between ____.

- a. 40-70 %
- b. 70-90 %
- c. 10-40 %
- d. All of the above

28. Type of rice husk furnace that burns the fuel while it is being spread in a circular pattern inside the combustion chamber.

- a. Step-grate furnace
- b. Suspended-air furnace
- c. Cyclone furnace
- d. All of the above

29. A biomass furnace is to be designed to provide heat for steaming lemon grass. The furnace combustion chamber is expected to operate at 700°C. Which of the following would you

recommend as material for the wall of the furnace combustion chamber?

- a. 10mm-thick mild steel
- b. 10mm-thick boiler plate
- c. 10mm-thick stainless steel sheet
- d. None of the above

30. Burning efficiency of rice hull furnaces is between ____.

- a. 70-80 %
- b. between 80-90 %
- c. 90 and less than 100 %
- d. None of the above

31. A furnace delivers 2,000 kcal per hour of heat. What is the equivalent heat produced in kW?

- a. 1.5 kW
- b. 2.4 kW
- c. 3.1 kW
- d. None of the above

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$$2000 \frac{\text{kcal}}{\text{hr}} \times \frac{0.001234 \text{ kW}}{1 \text{ kcal/hr}} = 2.468 \text{ kW}$$

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Solution:

$$\begin{aligned} Q_h &= m c_p (T_s - T_1) \\ &= 2 \text{ m}^3 \times 1000 \text{ kg/m}^3 \times 1 \text{ kcal/kg-}^\circ\text{C} \times (100^\circ\text{C} - 27^\circ\text{C}) \\ &= 146 \text{ kcal / 1 hr} \\ &= 146,000 \text{ kcal/hr} \end{aligned}$$

$$\begin{aligned} Q_e &= m \times h_{fg} \\ &= 100 \text{ kg water /hr} \times 540 \text{ kcal/kg} \\ &= 54,000 \text{ kcal/hr} \end{aligned}$$

$$\begin{aligned} \text{Pi heating} &= 146,000 \text{ kcal} / (0.4 \times 0.7) \\ &= 521,428.6 \text{ kcal/hr} \times 0.0012 \text{ kW-hr/kcal} \\ &= 625.7 \text{ kWt} \end{aligned}$$

$$\begin{aligned} \text{Pi evaporating} &= 54,000 \text{ kcal/hr} / (0.4 \times 0.7) \\ &= 192,857.1 \text{ kcal/hr} \times 0.0012 \text{ kW-hr/kcal} \\ &= 231.4 \text{ kWt} \end{aligned}$$

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32. Type of steam boiler commonly used in food industry.

- a. Fire-tube boiler
- b. Water-tube boiler
- c. Power boiler
- d. None of the above

33. Fuel blocks produced through the process of compaction with no binder instead the coherence is created by applying high temperature to the material and creating a hardened external shell.

- a. Densified briquettes
- b. Char briquettes
- c. Charcoal briquettes
- d. None of the above

34. Heat energy content of coconut shell charcoal is 14223 BTU/lb. How many kilo calories are there in 10 kilos of coconut shell charcoal.

- a. 78, 852.3 kcal
- b. 68, 852.3 kcal
- c. 87, 852.3 kcal
- d. None of the above

35. Organic matter that develops as a result of incomplete decomposition of wetland vegetation under conditions of excess moisture and oxygen deficiency.

- a. Compost
- b. Sludge
- c. Peat
- d. None of the above

36. Fuel blocks produced through simple compaction of biomass usually plants or plant residues in combination with binding material.

- a. Densified briquettes
- b. Char briquettes
- c. Charcoal briquettes
- d. None of the above

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37. Typical overall thermal efficiency of rice husk power plant using steam boiler and steam turbine for generating electricity.

- a. 15 to 30%
- b. 30 to 60%
- c. 60 to 80%
- d. None of the above

38. Typical specific power input of rice husk power plant.

- a. 1 to 2 t/hr-MW
- b. 2 to 4 t/hr-MW
- c. 4 to 6 t/hr-MW
- d. None of the above

39. Commonly used grate system for rice husk power plant.

- a. Step-grate
- b. Traveling grate
- c. Inclined grate
- d. None of the above

40. Factors that affect the overall thermal efficiency of power plant.

- a. Moisture content
- b. Combustion technology
- c. Plant design
- d. All of the above
- e. Two of the above

41. The amount of air needed to completely burn biomass fuel turning its by-product into ash.

- a. Air-fuel ratio
- b. Stoichiometric air
- c. Secondary air
- d. All of the above

42. Air that is required to directly burn fuel for combustion.

- a. Primary air
- b. Secondary air
- c. Tertiary air
- d. All of the above

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$$SA = \frac{2.67C + 8H + 1.5 - O}{0.23}$$

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43. By-products of combustion of biomass in a direct-fired furnace.

- a. CO₂
- b. CO
- c. Water vapor
- d. NO_x - nitrogen
- e. SO_x - sulfur
- f. Soot
- g. Fly ash
- h. All of the above
- i. Four the above

44. Another term for "soot."

- a. Fly ash
- b. Black carbon
- c. Smoke
- d. None of the above

45. The cooking heat requirement per person.

- a. 0.25 kWt - thermal
- b. 0.50 kWt
- c. 0.75 kWt
- d. None of the above

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GOD BLESS !!!

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